

Increasing functional cure rates after nucleo(s)tide analogue withdrawal: Is peg-IFN the answer?

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Nucleo(s)tide analogues (NAs) are well-tolerated and highly effective in suppressing HBV DNA in patients with chronic hepatitis B (CHB). Unfortunately, the risk of hepatocellular carcinoma (HCC) may persist, particularly in patients who do not achieve functional cure, defined as loss of HBsAg. As functional cure is rarely achieved with NA monotherapy,¹ alternative strategies are being explored to increase HBsAg clearance.

Recent trials with novel compounds suggest that immune modulatory agents likely play a key role in the pursuit of achieving functional cure.^{2,3} In particular, pegylated interferon (peg-IFN) is re-emerging in HBV cure studies because it has demonstrated efficacy in the past and, unlike new immune modulatory compounds, we know how to handle its toxicity profile which appears less prominent in HBV than HCV. In addition, peg-IFN was approved decades ago and is therefore available and affordable.

As the ultimate goal remains off-treatment functional cure, recent trials often include study arms where patients discontinue NA therapies. Interestingly, NA withdrawal itself has been found to have certain immunomodulating effects as well. Multiple studies^{4–8} have shown that withdrawal of NA therapy leads to increased HBsAg clearance compared to continued treatment. Although the mechanisms behind these observations are still unclear, it is hypothesized that the viral rebound following treatment cessation stimulates the immune response.⁹ As a result, NA therapy withdrawal has gained attention as another potential strategy for achieving functional cure.

NA withdrawal is however not without any risks, since viral relapses may also trigger an excessive immune response resulting in severe hepatitis, potentially leading to liver failure and even death.^{10,11} This necessitates close monitoring and clear re-treatment strategies to lower the risk of adverse outcomes.^{5,12} Thus, the decision to stop NA therapy should balance the potential benefits

(HBsAg loss) against the risks (severe flares and decompensation). Multiple large studies^{4,12–14} have identified several determinants of post-treatment outcomes. Factors associated with better outcomes are low serum HBsAg levels, undetectable levels of hepatitis B core-related antigen and/or HBV RNA, lower pretreatment HBV DNA levels, and HBV genotype C among Asian patients.^{15–17} The number of patients meeting these criteria is low, and even among those with the most favorable characteristics, not all will achieve HBsAg loss and severe flares can still occur.¹²

Since it is hypothesized that re-invigoration of the immune response plays an important role in determining the outcomes after NA cessation, introduction of peg-IFN as a second immune-modulating modality after NA withdrawal seems a promising strategy.¹⁸ Peg-IFN interacts with a broad variety of immune cells (HBV-specific T cells, B cells and natural killers cells) and inhibits viral replication,¹⁹ which can both potentially improve outcomes after NA withdrawal by reducing or delaying relapses and increasing HBsAg clearance.

In the current issue of the Journal, Li *et al.* report on the results of a randomized-controlled trial exploring the role of switching to peg-IFN in patients considered eligible for withdrawal of NA therapy.²⁰ In this study, 180 HBeAg-negative patients with CHB without advanced fibrosis who were treated with NA therapy for at least 2.5 years and virally suppressed for at least 1.5 years were randomized to either discontinue NA therapy, or to switch to peg-IFN for 48 weeks. The primary outcome was the virological relapse rate during 96 weeks after cessation of NA therapy, with HBsAg loss being a secondary outcome.

At baseline, groups were balanced with regards to HBsAg levels (median 2.70 vs. 2.73 log₁₀ IU/ml) and other potential confounders. At 96 weeks after NA withdrawal, the viral relapse rate was significantly lower in patients randomized to peg-IFN compared

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to the control group (21% vs. 54%, $p < 0.001$), whereas HBsAg loss rates were higher (22% vs. 9%, $p = 0.03$). As expected, lower baseline HBsAg levels were associated with more HBsAg loss and lower viral relapse rates. Peg-IFN did reduce the risk of relapse even further, also among patients with HBsAg < 100 IU/ml (11% for the peg-IFN vs. 24% in NA-stop group). The tolerability of peg-IFN in the current study is in line with previous reports, with the majority of patients experiencing the characteristic flu-like syndrome and fatigue, but the risk of severe adverse events was low.

Importantly, although immune stimulation as induced by peg-IFN might be expected to increase the risk of severe hepatitis flares among patients experiencing viral rebound, this was not the case. If anything, the risk of severe flares was lower among patients randomized to the peg-IFN switch (2.2 vs. 4.4%).

There are some limitations in this interesting study that should be taken into consideration. The primary and secondary endpoints were assessed at 96 weeks after NA cessation, which means that the control group had been free from treatment for 96 weeks, whereas the patients randomized to peg-IFN were off-treatment for only 48 weeks (as they were on peg-IFN for the first 48 weeks after NA withdrawal). As peg-IFN also has direct antiviral effects, the reduced risk of viral relapse during the first 48 weeks in this group is to be expected. The viral relapse rate at 48 weeks after withdrawal of all treatment (*i.e.* week 48 for the control group and week 96 for the peg-IFN group) favored the peg-IFN group (17.8% vs. 36.7%, $p = 0.007$). It thus cannot be ruled out that peg-IFN therapy may have merely delayed relapse in a subset of patients. Longer off-treatment follow-up is therefore required to confirm the robustness of these findings. The lack of longer follow-up is also relevant in the context of the significant risk of seroreversion that occurred in patients who experienced HBsAg loss with peg-IFN (12% for the peg-IFN vs. 0% in the NA-stop group). This observation is consistent with the results reported from a randomized trial that evaluated the efficacy of peg-IFN add-on to tenofovir, underscoring that the use of cumulative HBsAg loss rate is not always a valid endpoint.^{21,22}

Another question that should still be answered is whether the moment of starting peg-IFN has an impact on the outcomes as well. It can be argued that 'switching' results in similar outcomes compared to the 'add-on' strategy, because of the direct inhibitory effects of peg-IFN on viral replication (Table 1).¹⁹ Since reappearance of the virus is thought to be needed for re-activation of the host immune system,²³ the

inhibition of viral replication by peg-IFN after NA withdrawal may be similar to the viral suppression achieved with NAs during peg-IFN add-on therapy. This concept is supported by the SWAP-trial,²⁴ in which switching to peg-IFN resulted in more pronounced HBsAg decline and more HBsAg clearance compared to continuation of NA therapy. Whereas no differences were seen between the switch- and add-on group (12% vs. 13%, $p = 0.90$). To boost the immune restoration triggered by viral relapses, it can be suggested to start peg-IFN during these viral or even clinical relapses.^{18,25} However, caution should be taken that these relapses do not further deteriorate into severe flares by overstimulation of the immune system.¹⁰ No randomized-controlled trial has been performed to further evaluate this concept and thus further careful research focusing on enhancing the immune response after NA withdrawal with peg-IFN or other immune modulators can be considered.

Lastly, it remains unclear which patients would benefit the most from a strategy of switching from NA to peg-IFN. Since end-of-treatment HBsAg levels are the most important predictor of outcomes after NA withdrawal, it makes sense to assess the benefit of peg-IFN across HBsAg level strata. Based on previous studies, the chance of HBsAg clearance after NA withdrawal is thought to be negligible in patients with end-of-treatment HBsAg $> 1,000$ IU/ml, and this group therefore has the largest unmet need. Unfortunately, in the current study, the number of patients with HBsAg loss within this population was low despite peg-IFN switch (7.1% if HBsAg 1,000–1,500, 0% if HBsAg $> 1,500$ IU/ml). Among patients with slightly lower HBsAg levels, between 100–1,000 IU/ml, HBsAg loss was achieved in 14% in the peg-IFN switch arm vs. 6% in the control arm, suggesting a limited benefit in this subgroup. Patients with HBsAg < 10 IU/ml had extremely high HBsAg loss rates after peg-IFN switch of 100%, but also did well without peg-IFN (50% HBsAg loss) and are expected to achieve even higher HBsAg loss rates with prolonged follow-up.⁷ Finally, HBsAg loss was observed in 37.5% of patients with HBsAg 10–100 IU/ml when allocated to peg-IFN switch, compared to 0% in the control arm. Taken together, these findings identify patients with HBsAg 10–100 IU/ml at end of NA treatment as the group of patients most likely to benefit from switching to peg-IFN. Unfortunately, no data are available on the outcomes stratified by other predictors of peg-IFN response, such as HBV genotype.

In conclusion, the study by Li *et al.* provides valuable new information on the efficacy of peg-IFN in reducing relapse and

Table 1. Overview of outcomes of using NA and peg-IFN in HBeAg negative patients with chronic hepatitis B.

Type of therapy	HBsAg loss rates*	Risk of flares	Medication related side effects
NA monotherapy	0–2% (2 years) ¹	-	+/-
NA withdrawal	4–10% (2 years) ^{4–7}	+++	-
Peg-IFN switch	10–12% (1.5 years) ²⁴	++	+++
Peg-IFN add-on	8–13% (2 years) ²¹	+	+++
Peg-IFN <i>de novo</i>	5–6% (2 years) ²⁶	++	+++

NA, nucleos(t)ide analogues; peg-IFN, pegylated interferon.

Differences in patient characteristics between cohorts and types of therapy, for instance ethnicity and genotype, must be considered when interpreting the results above, as these variations may influence certain outcomes. Cited references do not encompass all available studies.

*Time (years) since start of Peg-IFN/NA or NA withdrawal.

increasing HBsAg loss rates in a subset of patients who discontinue NA therapy. The study results are also an important proof of concept supporting future trials in which peg-IFN is

used to increase or consolidate functional cure rates obtained with novel antivirals or immune-modifying compounds.

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Authors' contributions

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Supplementary data

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